Name:Vemula Srinidhi Date: 26-6-2026

Lab 3: Advanced Hyperparameter Optimization (CO2)

Key Concepts

Grid Search: An exhaustive search through a specified subset of hyperparameters. It is systematic but computationally expensive.

Random Search: A technique where random combinations of hyperparameters are selected. It is often more efficient than Grid Search for finding optimal values in high-dimensional spaces.

Generalization: The ability of a model to perform accurately on new, unseen data, rather than just 'memorizing' the training set.

Part 1: Theoretical Analysis

Before running your optimization scripts, analyze the theoretical impact of specific hyperparameters on model behavior. Consider how these parameters influence the trade-off between underfitting and overfitting.

1. **Impact of Batch Size:** How does the choice of batch size (e.g., 16 vs. 256) affect the convergence speed and the stochastic nature of the gradient descent path? How does it relate to the model's ability to generalize to the test set?

Batch size determines the number of training samples processed before updating the model weights. A **small batch size (e.g., 16)** performs more frequent weight updates, making the gradient descent path more stochastic (noisy). This often improves the model's ability to generalize to unseen test data but increases the training time. A **large batch size (e.g., 256)** provides smoother and faster computation because fewer updates are performed, but it may converge to a less optimal solution and reduce generalization performance, increasing the risk of overfitting.

2. **Role of Dropout Rate:** Explain the mechanism by which dropout acts as a regularization technique. What happens to the model's capacity and generalization error as the dropout rate increases from 0.0 to 0.5?

Dropout is a regularization technique that randomly disables a fraction of neurons during each training iteration. This prevents the network from relying too heavily on specific neurons and reduces overfitting. A **dropout rate of 0.0** means no neurons are dropped, which may lead to overfitting. Increasing the dropout rate to **0.5** improves generalization by forcing the network to learn more robust features. However, an excessively high dropout rate may reduce the model's learning capacity and lead to underfitting.

Part 2: Output Analysis & Optimization Log

Execute a Grid Search or Random Search for your MLP model. You must test at least **five different combinations** of hyperparameters (e.g., varying Learning Rate, Hidden Units, Batch Size, or Dropout). Record the results below to identify the optimal configuration.

Trial	Hyperparameters	Training Loss	Validation Accuracy	Overfitting?
1	LR=0.001, Hidden=128, Batch=32, Dropout=0.2	0.0812	97.32	no
2	LR=0.001, Hidden=256, Batch=64, Dropout=0.3	0.0769	97.68	no
3	LR=0.0005, Hidden=256, Batch=128, Dropout=0.5	0.1649	96.47	no
4	LR=0.01, Hidden=128, Batch=64, Dropout=0.2	0.1554	96.28	no
5	LR=0.001, Hidden=512, Batch=32, Dropout=0.4	0.0667	97.68	no

Optimal Configuration Summary: Identify the best performing set of hyperparameters from your trials. Justify why this specific combination provided the best balance between training performance and validation generalization.

The best performing configuration was **Learning Rate = 0.001, Hidden Units = 512, Batch Size = 32, and Dropout = 0.4**. This combination achieved the highest validation accuracy with a low training loss while avoiding overfitting. A moderate learning rate ensured stable convergence, the larger hidden layer improved feature learning, the smaller batch size enhanced generalization, and dropout reduced overfitting by preventing excessive dependence on individual neurons.